FINAL REPORT

MAT-3991

Group 05

TEAM MEMBERS

- S/20/832 - W.H.J. Induwara
- S/20/816 - H.G.U.P.Dilhara
- S/20/835 - W.A.D.J.S.Jayasena
- S/20/858 - M.S.S.Rasandika
- S/20/867 - H.L. Samarawickrama
- S/20/885 - L.B.Lamahewa
- S/20/811 - W.M.A.U Bandara
- S/20/820 - E.M.K.C.Ekanayake

ABSTRACT

This project addresses the industrial challenge of foot warpage in porcelain cup manufacturing, undertaken in collaboration with Noritake Lanka Porcelain Private Limited. The primary objective was to identify, quantify, and model the relationship between key process variables and the final 'warp percentage' defect rate.

Process data was collected for variables including clay hardness, clay moisture, drying temperatures, and unloading moisture levels at two different stages. Due to production scheduling constraints, the initial real-world dataset was limited. To demonstrate the viability of the proposed analytical workflow, an expanded dataset was simulated based on company-specified standard ranges.

A Beta Regression model was selected as the most statistically appropriate method for modeling the response variable, which is a continuous proportion bounded between 0 and 1. This approach correctly handles the non-normal distribution of the proportional data. The final model, which included main effects and a key two-way interaction, demonstrated strong predictive performance (Test $R^2 = 0.7718$, Test RMSE = 0.0435).

The analysis identified clay hardness, unloading moisture 1, unloading moisture 2, and a significant interaction between clay moisture and drying temperature 1 as the most critical factors influencing foot warpage. This report concludes with a set of preliminary, quantitative recommendations for process adjustments to minimize defect rates. The study successfully provides a robust statistical framework that can be applied to a larger, future dataset to validate and refine these findings for implementation.

ACKNOWLEDGEMENT

We would like to express our deepest appreciation to our supervisors and evaluators, whose continuous guidance, constructive feedback and encouragement were instrumental in shaping this project from its initial concept to its final implementation. Their expertise helped us maintain technical accuracy and industry relevance at every stage.

We would also like to extend our sincere gratitude to the management and staff of Noritake Lanka Porcelain Private Limited for providing us with this invaluable opportunity. Their cooperation, insights into the porcelain manufacturing process, and support during our data collection were essential to the completion of this study.

This project is the result of collective effort, and we acknowledge with appreciation every individual who contributed to its success.

CONTENT

Team members I
Abstract II
Acknowledgement III
1. Introduction 1
    1.1 Aim and motivation 1
    1.1 Company overview 1
2. Objectives 1
3. Data Collection 2
    3.1 Data Collection Process 2
    3.2 Challenges In Data Collection 2
4. Dataset 3
5. Data Analysis 4
6. Feature Engineering 6
7. Model Selection 7
8. Model Assumptions 7
9. Model Fitting And Evaluation 8
    9.1 Overview of beta regression model 8
    9.2 Data splitting and initial mean model selection 8
    9.3 Precision model assessment 9
    9.4 Interaction effect identification and final model selection 9
    9.5 Model validation and performance metrics 9
    9.6 Model evaluation 10
10. Residual Analysis 11
11. Conclusions 15
12. References 16

1. INTRODUCTION

1.1 Aim And Motivation

The aim of this to bridge academic learning with real-world industrial challenges by applying quantitative techniques, analytical thinking, and mathematical modeling. This not only provides us with valuable practical exposure but also allows us to contribute to the improvement of industrial operations in a real setting.

1.2 Company Overview

Noritake Lanka Porcelain Private Limited is the pioneer porcelain tableware manufacturing conglomerate in Sri Lanka. The Company was established in 1972 as a joint venture company between Ceylon Ceramic Corporation and Noritake Co., Limited of Japan. Representing the Noritake global family in Sri Lanka, the Company has evolved into Noritake Lanka Porcelain Private limited, a new giant in the porcelain industry, earning a reputation in an international market for standard and quality. Noritake began manufacturing Western style tableware for export in 1973. For over 47 years, they have delivered tableware to dining tables in every corner of the world

2. OBJECTIVES

Objective 01

To study the relationship between clay properties and foot warpage.

Objective 02

To use statistical analysis (e.g., correlation, regression) to identify the most significant variables contributing to warpage.

Objective 03

To develop a predictive model or decision framework to control foot warpage in the jiggering process.

3. DATA COLLECTION

3.1 Data Collection Process

In our project, we aim to identify the causes of foot warpage in porcelain cups and explore ways to control it. To achieve this, we have collected data at various stages of the cup-making process.

- **Clay Sample Analysis**:

For each batch of clay, we took a small sample and measured its hardness using a Durometer. This helps us determine the hardness or softness of the clay before it's shaped.

- **Moisture Content Measurement:**

Next, we assessed the moisture content of the clay. We extracted 100 grams from the center of a clay sample, ensuring it remained moist, and placed it in an oven for about one hour. We then recorded its weight to calculate the moisture content using a specific formula.

- **Jiggering Process Observations:**

During the initial drying phase after jiggering, we recorded the first drying temperature five minutes into the process. Once this stage concluded, we measured the first unloading moisture of a cup using the same moisture calculation method.

- **Second Drying Stage Measurements:**

Prior to the second drying phase, we noted the second drying temperature. After completing this stage, we measured the second unloading moisture of a cup, again applying the same moisture calculation method.

- **Firing Department Results:**

Once the forming process is finalized, the cups are transferred to the firing department. After approximately one week, we received the warp percentages for each batch of cups.

This outlines our step-by-step data collection process.

3.2 Challenges in Data Collection

- **Production Scheduling Constraints**

The primary challenge we faced at the outset was gaining access to production runs. Because of the existing inventory of porcelain cups, the facility couldn't dedicate production capacity exclusively for data collection. This situation required a waiting period until regular commercial production could resume, ultimately delaying the commencement of the data collection phase.

- **Data Quality and Inconsistency**

During the active data collection phase, inconsistencies in the manufacturing process resulted in the gathering of unreliable data points. As a result, some of the collected dataset needed to be discarded to uphold the integrity and quality of the training data. This selective removal was essential to ensure that the model was trained solely on data reflecting a consistent and valid production state.

4. DATASET

Due to the limited dataset containing only seven observations, we needed to create an expanded dataset using a generative AI tool. We provided the original dataset along with standard ranges for each variable as specified by the company. It's important to note that the simulation of data is outside the scope of our project; it was performed solely to illustrate our workflow.

Lot NO:	Clay Hardness	Clay Moisture (%)	Drying Temperature 1(°C)		Unloading Moisture 1 (%)	Drying Temperature 2(°C)	Unloading Moisture 2 (%)	Warp Percentage
			Analog	Digital				
1	83	23	43	44.5	18.56749311	64.5	5.705705706	0.416666667
2	84	23	46	44.5	18.6345831	65.5	5.132075472	0.363636364
3	81	23.1	46	45	18.65536039	66	5.538922156	0.181818182
4	86	22.6	46	45	18.7426557	67.5	4.744255004	0.1
5	85	22.6	46	45	18.42709529	68	5.19005848	0.454545455
6	89	22.5	46	45	18.44262295	68.5	4.966641957	0.25
7	84	22.5	47	45	18.23416507	69.5	5.487804878	0.181818182

table 1: The actual collected dataset

5. DATA ANALYSIS

In this section, we examined the previously mentioned dataset for any significant patterns or anomalies. We found no missing values, and the summary statistics confirmed that all variables fall within reasonable and expected ranges.

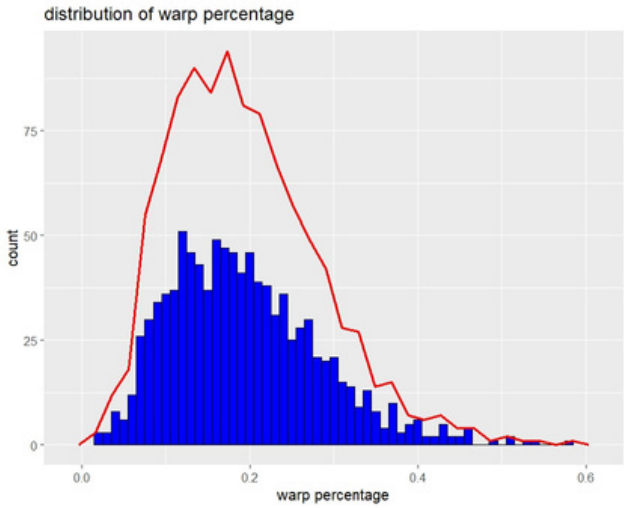

figure 1: Distribution of the response variable

The *figure 1* indicates that the response variable, which is the warp percentage, appears to follow a beta distribution.

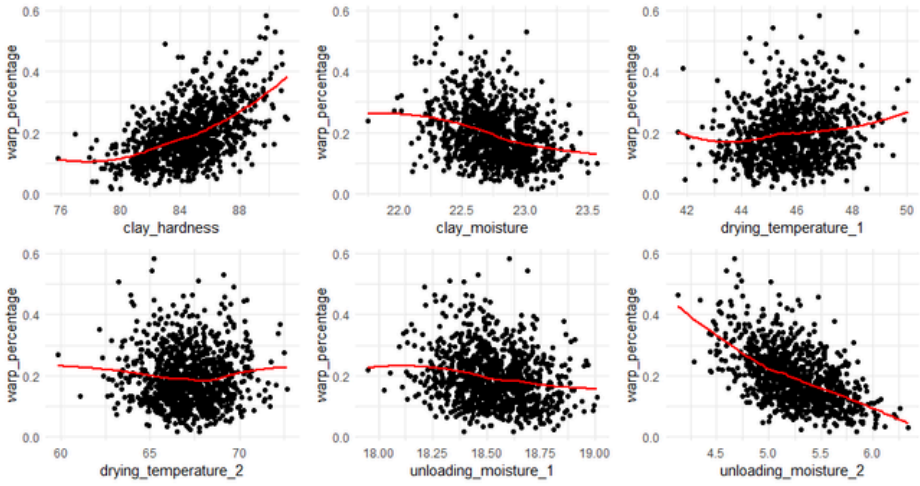

figure 2: Scatter plots of dependent variable against each independent variable

From the scatter plots illustrated in figure 2, we can observe distinct relationships between certain independent variables and the response variable. This observation encourages us to explore the possibility of a regression model.

As illustrated in *figure 3* below, there appears to be a correlation between certain independent variables and the dependent variable. Additionally, the lack of significant correlation with the dependent variable indicates that severe multicollinearity is not present.

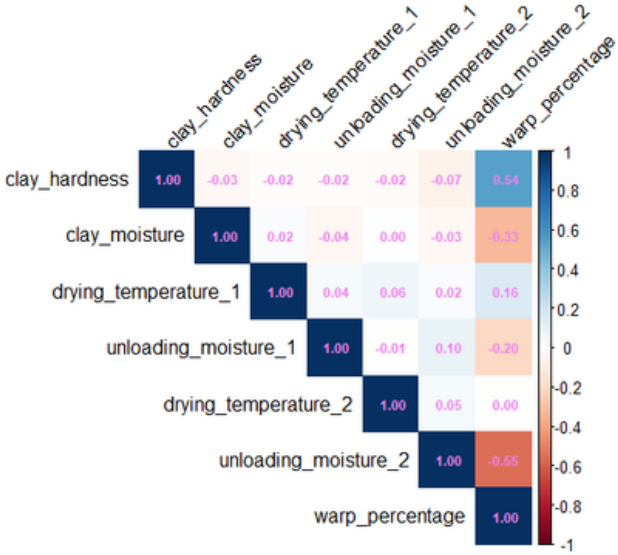

figure 3: Correlation heatmap

To verify the absence of multicollinearity, we calculated the Variance Inflation Factor (VIF) for each independent variable.

Variable	**VIF (Variance Inflation Factor)**
clay_hardness	1.006280
clay_moisture	1.004388
drying_temperature_1	1.006508
unloading_moisture_1	1.012944
drying_temperature_2	1.006932
unloading_moisture_2	1.017025

table 2: Variance inflation factor for independent variables

As indicated in the *table 2*, the VIF values are significantly below the cutoff threshold of 10. Thus, we can confidently conclude that multicollinearity is not present.

6. FEATURE ENGINEERING

From the original independent variables, we attempted to derive new features to determine if they would enhance our analysis.

The newly derived features are:

- Drying temperature difference
- Unloading Moisture difference
- Total Moisture loss

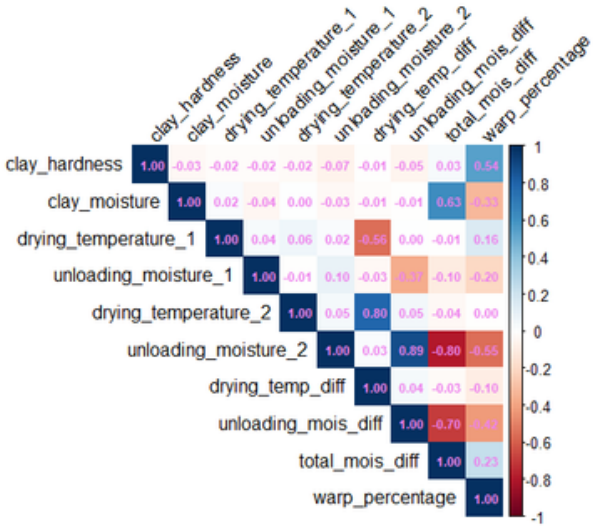

figure 4: Correlation heatmap with new features

From *figure 4*, it is evident that the introduction of new features leads to multicollinearity, which violates a fundamental assumption of our chosen model. Consequently, we decided to eliminate these features and retained only the original variables.

7. MODEL SELECTION

We explored multiple regression models, including Poisson, negative binomial, logistic regression, and simple linear regression. Our initial intention was to use the number of warped cups as the response variable. However, the large lot sizes made it challenging for the production team to provide accurate count data. Instead, they supplied the warpage percentage for each batch.

Since warpage percentage is a continuous proportion ranging from 0 to 1, the Poisson and negative binomial regression models were deemed unsuitable. We then considered a simple linear regression model, but this approach could yield unrealistic predictions outside the (0,1) interval, such as negative proportions or values exceeding 1, leading to its rejection.

Ultimately, we assessed logistic regression; however, it is only applicable for binary outcomes, rendering it inappropriate for our continuous proportion response variable.

As a result, we chose to implement a **beta regression** model specifically designed for analyzing continuous proportions that lie between 0 and 1. This method adeptly manages the inherent heteroscedasticity and asymmetry often observed in proportion data, ensuring that the predicted values remain within the acceptable range. By utilizing a suitable link function, like the logit link, the model allows us to relate the warpage percentage to various process variables, such as hardness, clay moisture, drying temperatures, and unloading moisture, while providing accurate and interpretable estimates of their effects.

$$logit(\mu_i) = ln(\frac{\mu_i}{1-\mu_i}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k$$

8. MODEL ASSUMPTIONS

- **Response Variable Distribution:** The conditional distribution of response variable Y, given the predictors, must follow a Beta distribution
- **Response Variable Range:** The response variable must be continuous and fall strictly within the open unit interval (0, 1)
- **Link Function:** The expected value of the response variable is related to the linear predictor through a link function (commonly the logit link).
- **Independent Observations:** The observations must be statistically independent.
- **Variance is not constant (i.e., Heteroscedasticity)**
- **Correct Specification of the Mean Model:** The functional form relating the link-transformed mean to the predictors must be correctly specified (e.g., no omitted variables, no multicollinearity).

9. MODEL FITTING AND EVALUATION

9.1 Overview of Beta regression model

- **Introduction to Beta Regression Parameters**

Beta regression is a powerful statistical tool, primarily because it models two main parameters: mean and precision. Understanding how these parameters function is essential to effectively building a model.

- **Mean Model**

The mean model predicts the average of the ith observation. In our case, it predicts the average warp percentage. The mean parameter essentially provides insight into the central tendency of the data.

- **Precision Model**

Precision, on the other hand, controls the variance of the dependent variable. A higher precision indicates a lower variance, meaning the data is tightly clustered around the mean. Conversely, lower precision results in higher variance, indicating that the data is more dispersed.

The interplay between mean and precision within beta regression is what renders it a robust choice for various modeling tasks. By understanding how these parameters work, we can better interpret and utilize the model for accurate predictions.

9.2 Data Splitting and Initial Mean Model Selection

The complete dataset was initially partitioned into a training set (80%) and a testing set (20%) to ensure a robust model evaluation on unseen data.

The mean model was developed on the training dataset using a combination of backward and forward selection methods. The aim was to identify the optimal set of explanatory variables that predict the mean response. Model performance during this stage was primarily assessed using the Akaike Information Criterion (AIC) and Pseudo R^2 values, leading to the selection of a preliminary best-fit model.

9.3 Precision Model Assessment

In beta regression, it is possible to model the precision (or variability) of the process, which can indicate if any variable makes the process more erratic or consistent. This assessment was performed to check if any explanatory variable had a significant impact on the model's precision. The analysis concluded that none of the variables significantly altered the process's precision, leading to the decision to proceed with the selected mean model structure.

9.4 Interaction Effect Identification and Final Model Selection

All possible two-way interaction effects between the explanatory variables were systematically investigated. This analysis revealed a significant interaction between Clay Moisture and Drying Temperature 1. The inclusion of this interaction term substantially improved the model's accuracy.
The final model was selected based on a comparison of AIC and Pseudo R^2 values, resulting in a model comprising four main effects and one significant two-way interaction effect.

9.5 Model Validation and Performance Metrics

To obtain a more reliable estimate of the model's accuracy, 10-fold cross-validation was applied to the training dataset. The model exhibited strong performance on the training data, yielding the following average metrics:

Average Pseudo R^2: 0.72255

Interpretation: Pseudo R2 compares the performance of the full model against the null model (model with no predictors). A value of 0.72255 suggests the model is a good fit and explains a substantial amount of the variation in the response variable.

Average Root Mean Square Error (RMSE): 0.0446

I**nterpretation:** RMSE measures the average magnitude of the errors; specifically, how far the model's predictions are from the actual values. A low value of 0.0446 indicates the model's predictions are generally close to the true outcomes.

```
Call:
betareg(formula = warp_percentage ~ clay_hardness + clay_moisture + drying_temperature_1 + unloading_moisture
_1 +
    unloading_moisture_2 + drying_temperature_1:clay_moisture, data = train_set, link = "logit")

Quantile residuals:
    Min      1Q  Median      3Q     Max
-3.4284 -0.6683  0.0124  0.6835  3.5298

Coefficients (mean model with logit link):
                                    Estimate Std. Error z value Pr(>|z|)
(Intercept)                       100.556589  33.566532   2.996  0.00274 **
clay_hardness                       0.125940   0.004499  27.996  < 2e-16 ***
clay_moisture                      -4.401580   1.476705  -2.981  0.00288 **
drying_temperature_1               -1.724490   0.731748  -2.357  0.01844 *
unloading_moisture_1               -0.624299   0.064184  -9.727  < 2e-16 ***
unloading_moisture_2               -0.937939   0.032241 -29.091  < 2e-16 ***
clay_moisture:drying_temperature_1  0.079562   0.032207   2.470  0.01350 *

Phi coefficients (precision model with identity link):
      Estimate Std. Error z value Pr(>|z|)
(phi)    74.19       3.70   20.05   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Type of estimator: ML (maximum likelihood)
Log-likelihood:  1389 on 8 Df
Pseudo R-squared: 0.7223
Number of iterations: 186 (BFGS) + 4 (Fisher scoring)
```

9.6 Model Evaluation

The final model was then evaluated on the completely unseen testing dataset to confirm its generalization ability.

Test R^2: 0.7718

I**nterpretation:** The standard R^2 of 0.77 on the test set indicates that 77% of the total variation in the outcome variable can be explained by the model.

Test RMSE: 0.04354

Interpretation: The test RMSE being lower than the training RMSE is a positive indication. It suggests the model is performing well on unseen data and is not overfit to the training dataset, demonstrating excellent generalization capability.

10. RESIDUAL ANALYSIS

we perform residual analysis to check whether the model fits the data well and to ensure the assumptions of beta regression are satisfied

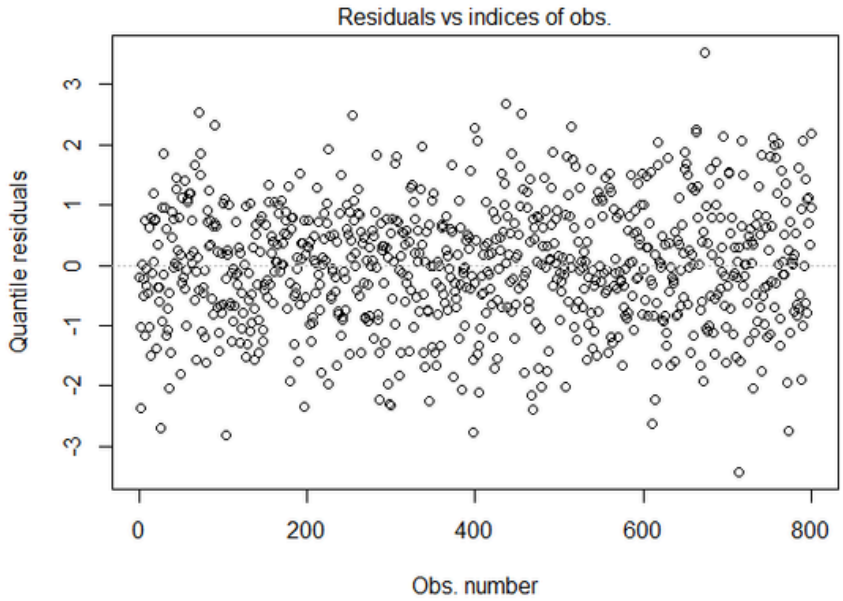

figure 5: Residuals vs indices of observations

In the *figure 5*, each point represents the residual for an individual observation. The residuals appear to be randomly scattered around zero, without any noticeable pattern, trend, or large spikes. This randomness indicates that the observations are independent, satisfying the independence assumption of the beta regression model.

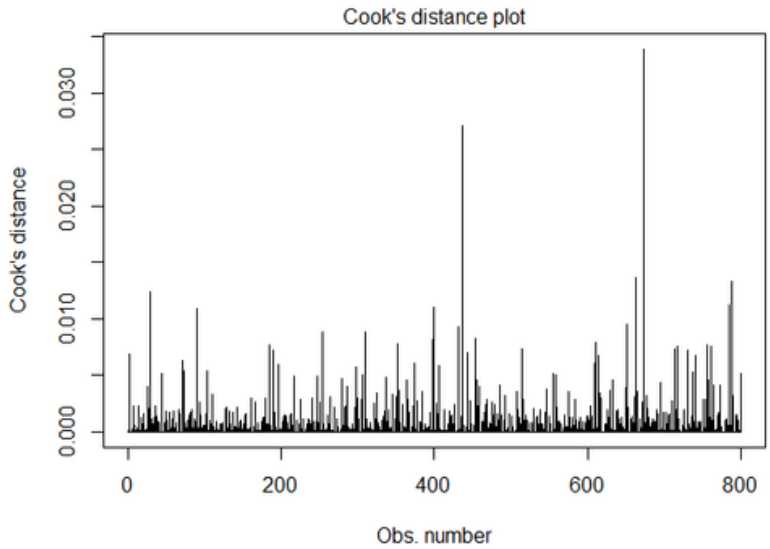

figure 6: Cook's distance plot

The *figure 6* represents the Cook's distance values for each observation. Except for two observations, all others have relatively low Cook's distance values. But since non of the cook's distance values are above the cutoff value of 0.898 (F(0.5,p,n-p)) there are no significant influential observations.

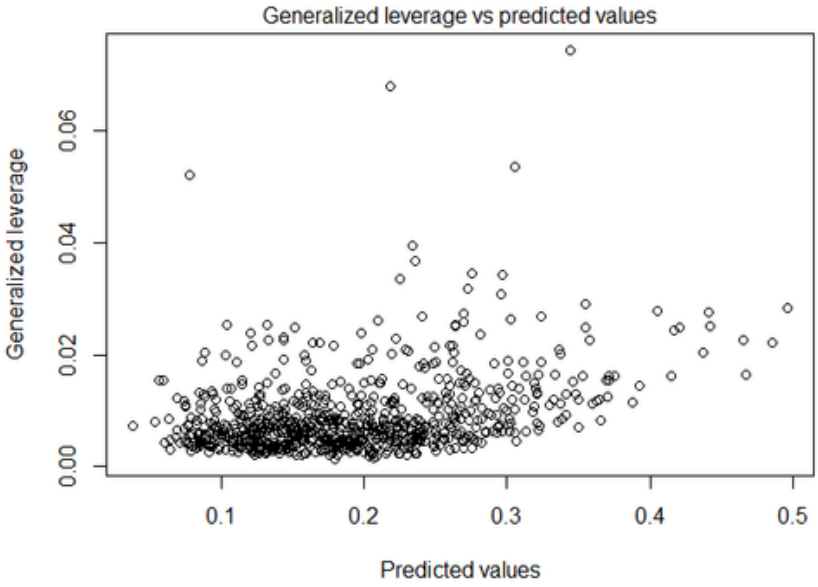

figure 7: Generalized leverage vs predicted values

The *figure 7* shows the generalized leverage values plotted against the predicted values. Most observations have low and similar leverage values, forming a horizontal band near the bottom of the plot. However, three observations exhibit relatively high leverage values, indicating that these points might have a stronger influence on the model fit compared to the others. But we from the cook's distance plot we confirmed that there are no significant influential observations

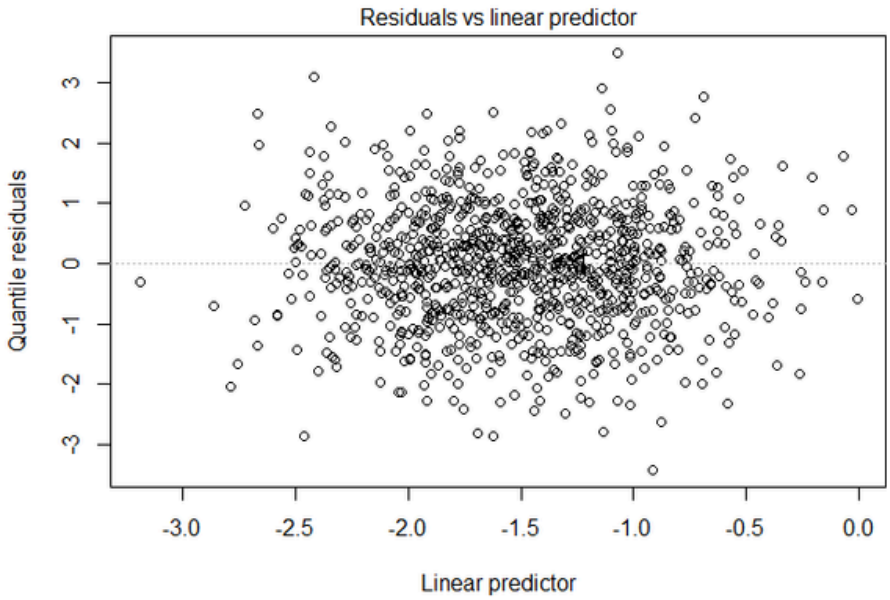

figure 8: Residuals vs linear pedictor

The *figure 8* represents the residuals plotted against the linear predictors. Most of the points are randomly scattered around zero, and no clear pattern or trend is observed. This indicates that the model is correctly specified with an appropriate link function.

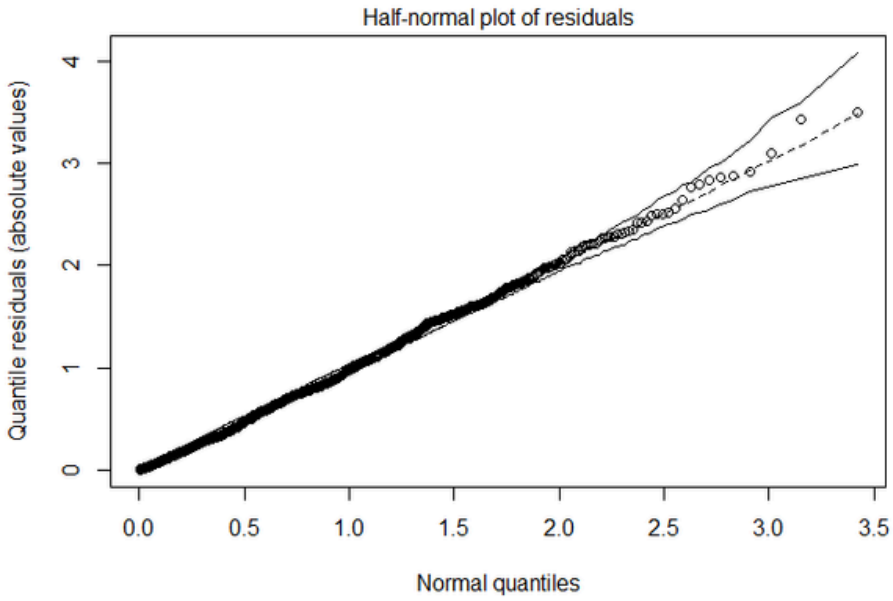

figure 9: Half-normal plot of residuals

The *figure 9* represents the half-normal plot of residuals against the normal quantiles. As observed, the points lie approximately along a straight line, indicating that the residuals follow a theoretical normal pattern and that the model fits the data well.

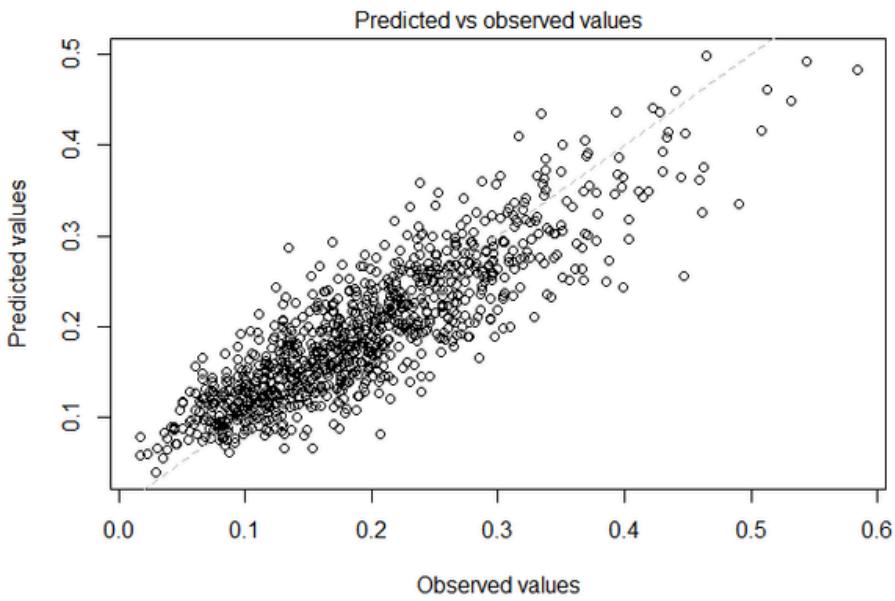

figure 10: predicted vs observed values

The *figure 10* shows the predicted values plotted against the observed values. The points are closely scattered around the 45-degree line, indicating that the beta regression model provides accurate predictions of the warpage percentage with minimal bias.

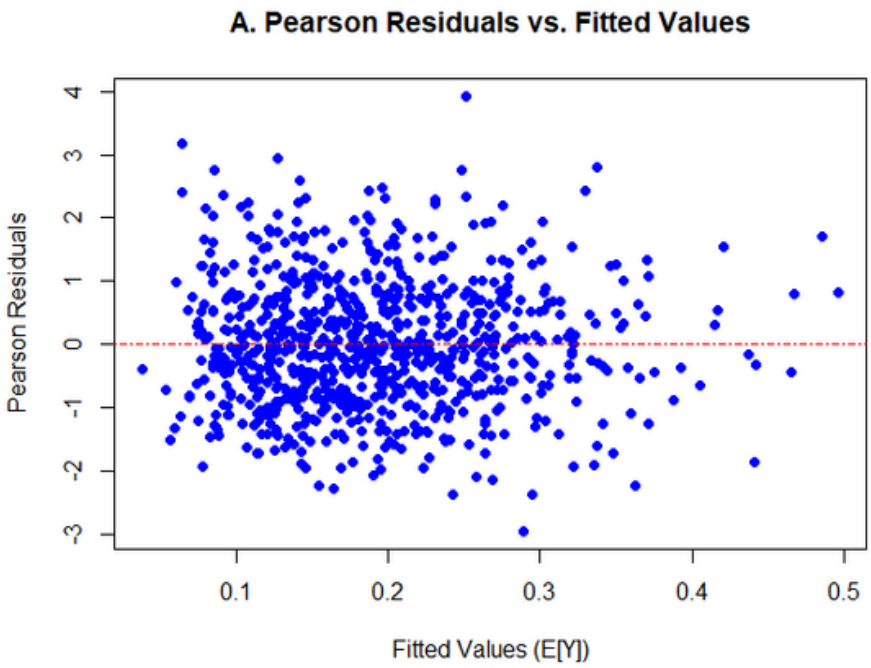

figure 11: Pearson residual vs fitted values

Since the variance depends on the mean in the beta regression model, heteroscedasticity is inherent to the model. Therefore, to obtain Pearson residuals against fitted values, the existing residuals were standardized. When plotting the Pearson residuals against the fitted values, the data points appear to be randomly scattered around zero without any noticeable pattern, trend, or change in spread. This indicates that the model is appropriately specified, linear, and satisfies the assumption of constant variance of the standardized residuals.

11. CONCLUSIONS

Based on the findings of our study, we recommend the following measures to improve the production process and reduce warpage percentage:

- Clay Hardness: Keep the clay hardness below 80 to minimize warping.
- First Unloading Moisture: Maintain the first unloading moisture above 20% to ensure proper material consistency.
- Second Unloading Moisture: Maintain the second unloading moisture above 10% to support dimensional stability during drying.
- Drying Temperature–Moisture Balance: Optimize the drying process by keeping moisture levels between 25–30% and drying temperatures below 50°C to reduce warpage.

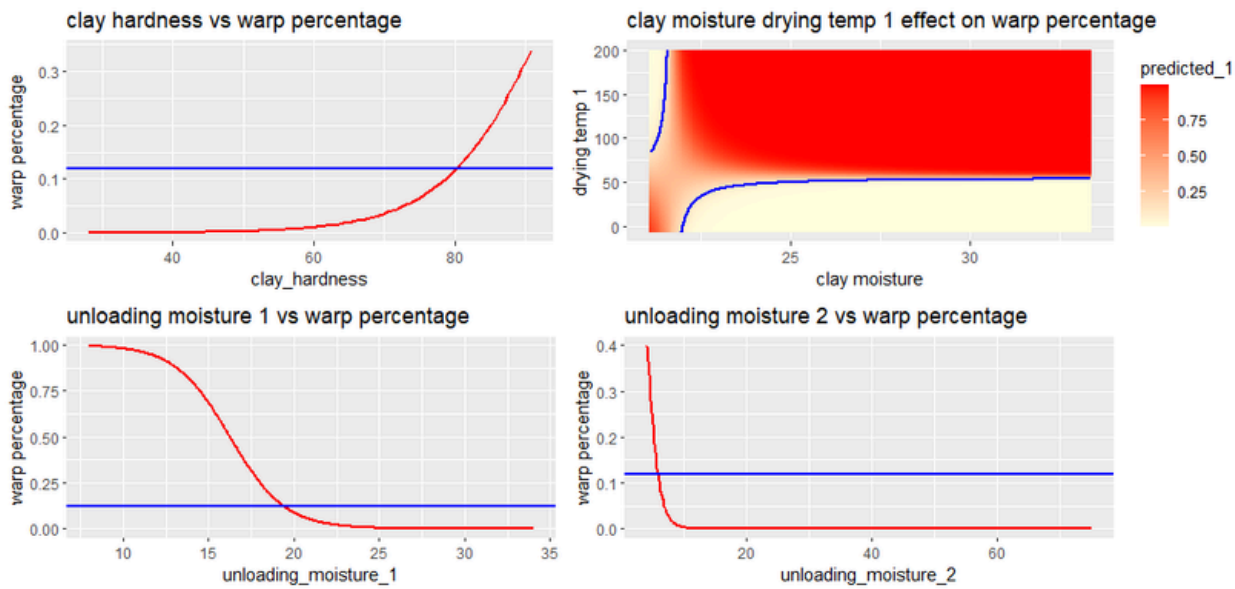

figure 12: Marginal effect plots

By following these recommendations, we believe the factory can achieve better accuracy, lower warp defect rates, and more improved process consistency.

The above recommendations are based on the Beta Regression model developed using the simulated dataset. While this report successfully demonstrates a valid workflow for identifying key variables, these findings are preliminary. The model and its conclusions must be validated with a much larger, real-world dataset collected from the production process before they can be implemented with confidence

12. REFERENCES

- Cribari-Neto, Francisco & Zeileis, Achim. (2010). Beta Regression in R. Journal of Statistical Software. 34. 1-24. 10.18637/jss.v034.i02.

- Espinheira, P. L., Ferrari, S. L. P., & Cribari-Neto, F. (2008). On beta regression residuals. Journal of Applied Statistics, 35(4), 407–419. https://doi.org/10.1080/02664760701834931

- Mangiafico, S.S. 2016. Summary and Analysis of Extension Program Evaluation in R, version 1.23.0, revised 2025. rcompanion.org/handbook/.